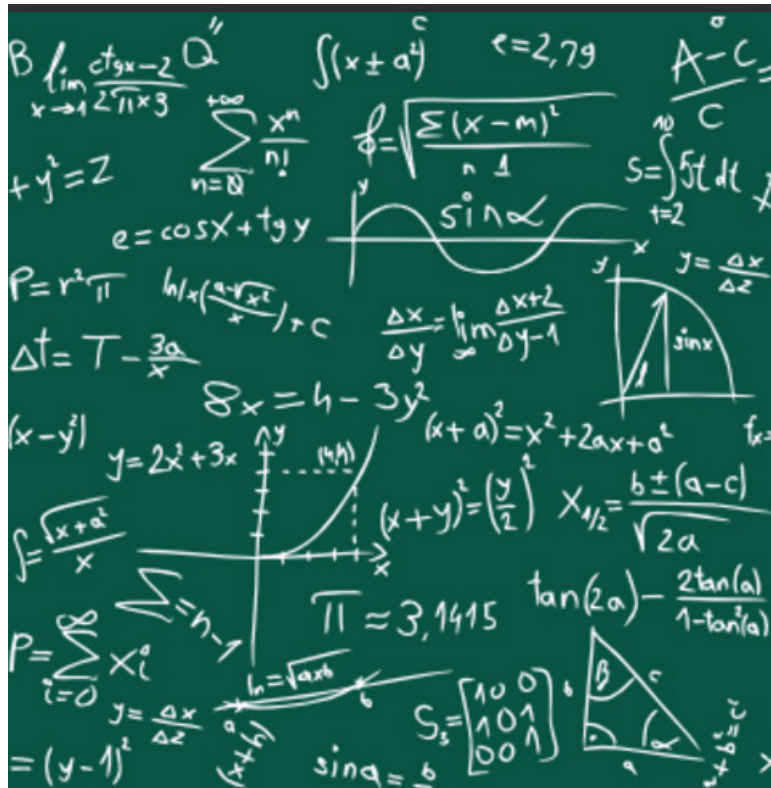


Exercises: principle of mathematical induction



- **Authors:** Fabio Furini and Gianluca Priori.
- **Book:** “Esercitazioni Analisi matematica 1”, Marco Bramanti, Esculapio.

Exercise 1

Prove by induction that for every integer $n \geq 1$ the following holds:

$$\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$$

(sum of the squares of the first n natural numbers)

By induction on n .

- **Base case**

Let $n = 1$. Then the statement becomes:

$$\sum_{k=1}^1 k^2 = \frac{1(1+1)(2 \cdot 1 + 1)}{6} \quad \text{i.e.} \quad 1 = 1$$

which is clearly true.

- **Inductive step**

Assume that it is true for n , and let us prove it for $(n+1)$. By the inductive hypothesis, we have:

$$\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}.$$

Hence we can write:

$$\begin{aligned} \sum_{k=1}^{n+1} k^2 &= \sum_{k=1}^n k^2 + (n+1)^2 \\ &= \frac{n(n+1)(2n+1)}{6} + (n+1)^2 = \frac{n(n+1)(2n+1) + 6(n+1)^2}{6} \\ &= \frac{(n+1)(n(2n+1) + 6(n+1))}{6} = \frac{(n+1)(2n^2 + n + 6n + 6)}{6} \\ &= \frac{(n+1)(2n^2 + 4n + 3n + 6)}{6} = \frac{(n+1)((2n)(n+2) + 3(n+2))}{6} \\ &= \frac{(n+1)(n+2)(2n+3)}{6} = \frac{(n+1)(n+1+1)(2(n+1)+1)}{6} \end{aligned}$$

which is exactly the desired statement for $n+1$.

Exercise 2

Prove by induction that for every integer $n \geq 1$ the following holds:

$$\sum_{k=0}^{n-1} (2k+1) = n^2 \quad \text{or} \quad \sum_{k=1}^n (2k-1) = n^2$$

(sum of the first n odd numbers)

By induction on n .

- **Base case**

Let $n = 1$. Then the statement becomes:

$$\sum_{k=0}^0 (2k+1) = 1^2 \quad \text{i.e.} \quad 1 = 1$$

which is clearly true.

- **Inductive step**

Assume that it is true for n , and let us prove it for $(n+1)$. By the inductive hypothesis, we have:

$$\sum_{k=0}^{n-1} (2k+1) = n^2$$

Hence we can write:

$$\begin{aligned} \sum_{k=0}^{n-1+1} (2k+1) &= \sum_{k=0}^{n-1} (2k+1) + 2n+1 \\ &= n^2 + 2n+1 \\ &= (n+1)^2 \end{aligned}$$

which is exactly the desired statement for $n+1$.

Exercise 3

Prove by induction that for every integer $n \geq 1$ the following holds:

$$\sum_{k=1}^n 2k = n(n+1)$$

(sum of the first n even numbers)

By induction on n .

- **Base case**

Let $n = 1$. Then the statement becomes:

$$\sum_{k=1}^1 2k = 1 \cdot 2 \quad \text{i.e.} \quad 2 = 2$$

which is clearly true.

- **Inductive step**

Assume that it is true for n , and let us prove it for $(n + 1)$. By the inductive hypothesis, we have:

$$\sum_{k=1}^n 2k = n(n + 1)$$

Hence we can write:

$$\begin{aligned} \sum_{k=1}^{n+1} 2k &= \sum_{k=1}^n 2k + 2(n + 1) \\ &= 2 \left(\sum_{k=1}^n k + (n + 1) \right) \\ &= 2 \left(\frac{n(n + 1)}{2} + (n + 1) \right) \\ &= 2 \left(\frac{n(n + 1) + 2(n + 1)}{2} \right) \\ &= (n + 1)(n + 2) = (n + 1)(n + 1 + 1) \end{aligned}$$

which is exactly the desired statement for $n + 1$.

Exercise 4

Prove by induction that for every integer $n \geq 1$ the following holds:

$$\sum_{k=1}^n \frac{k}{2^k} = 2 - \frac{n+2}{2^n}$$

By induction on n .

- **Base case**

Let $n = 1$. Then the statement becomes:

$$\sum_{k=1}^1 \frac{k}{2^k} = 2 - \frac{1+2}{2^1} \quad \text{i.e.} \quad \frac{1}{2} = \frac{1}{2}$$

which is clearly true.

- **Inductive step**

Assume that it is true for n , and let us prove it for $(n + 1)$. By the inductive hypothesis, we have:

$$\sum_{k=1}^n \frac{k}{2^k} = 2 - \frac{n+2}{2^n}$$

Hence we can write:

$$\begin{aligned} \sum_{k=1}^{n+1} \frac{k}{2^k} &= \sum_{k=1}^n \frac{k}{2^k} + \frac{n+1}{2^{n+1}} \\ &= 2 - \frac{n+2}{2^n} + \frac{n+1}{2^{n+1}} \\ &= 2 - \frac{2(n+2)}{2^{n+1}} + \frac{n+1}{2^{n+1}} \\ &= 2 - \left(\frac{2(n+2) - n - 1}{2^{n+1}} \right) \\ &= 2 - \left(\frac{n+3}{2^{n+1}} \right) \\ &= 2 - \frac{(n+1)+2}{2^{n+1}} \end{aligned}$$

which is exactly the desired statement for $n + 1$.

Exercise 5

Given a set X with n elements, prove by induction that the power set $\mathcal{P}(X)$ has 2^n elements, that is,

$$|\mathcal{P}(X)| = 2^n \quad \text{for } n \geq 0$$

By induction on n .

- **Base case**

Let $n = 0$. A set with 0 elements is the empty set ($\emptyset$). The power set of the empty set contains only the empty set as an element. Then the statement becomes:

$$|\mathcal{P}(\emptyset)| = 2^0 \quad \text{i.e.} \quad 1 = 1$$

which is clearly true.

- **Inductive step**

Assume the statement is true for a set of n elements, and let us prove it for a set of $(n + 1)$ elements. By the inductive hypothesis, the power set of a set of n elements has 2^n elements.

A set Y of $(n + 1)$ elements can be written as:

$$Y = \{a_1, a_2, \dots, a_n, a_{n+1}\}$$

hence:

$$\mathcal{P}(Y) = \{S : S \subseteq Y\} = \{S : S \subseteq \{a_1, a_2, \dots, a_n, a_{n+1}\}\}$$

The subsets of Y either contain the element a_{n+1} or do not contain it. We therefore define the set Z_a formed by the first n elements of Y :

$$Z_a = \{a_1, a_2, \dots, a_n\} \quad \text{hence} \quad |\mathcal{P}(Z_a)| = 2^n$$

We now define the following set Z_b :

$$Z_b = \{S \cup \{a_{n+1}\} : S \subseteq \underbrace{\{a_1, a_2, \dots, a_n\}}_{=Z_a}\} = \{S \cup \{a_{n+1}\} : S \subseteq Z_a\} \quad \text{and} \quad |Z_b| = 2^n$$

since Z_b has one element for each element of $\mathcal{P}(Z_a)$.

In other words, $\mathcal{P}(Z_a)$ contains all the subsets of Y in which the element a_{n+1} is not present, while Z_b contains all the subsets of Y in which the element a_{n+1} is present.

We also note that the sets $\mathcal{P}(Z_a)$ and Z_b are disjoint, that is: $\mathcal{P}(Z_a) \cap Z_b = \emptyset$. Hence $|\mathcal{P}(Z_a) \cup Z_b| = |\mathcal{P}(Z_a)| + |Z_b|$. Finally, we have:

$$|\mathcal{P}(Y)| = |\mathcal{P}(Z_a) \cup Z_b| = 2^n + 2^n = 2^{n+1}$$

which is exactly the desired statement for $n + 1$.